Fundamental rules of Probability

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Probability distributions follow the rules of probability

1. Non-negativity
2. Exhaustiveness
3. Addition Rule
4. Multiplication Rule
5. Complementary Events

Non-negativity

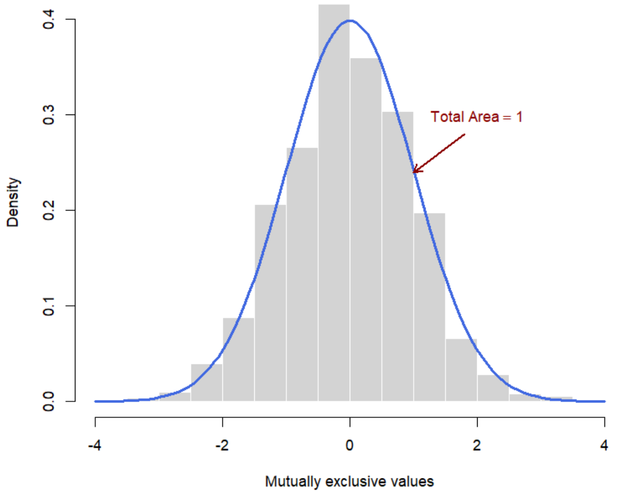
- The probability of any event E_i is assigned a non-negative number

$$P(E_i) \geq 0$$

Exhaustiveness

- The sum of the probabilities of all mutually exclusive outcomes is equal to 1

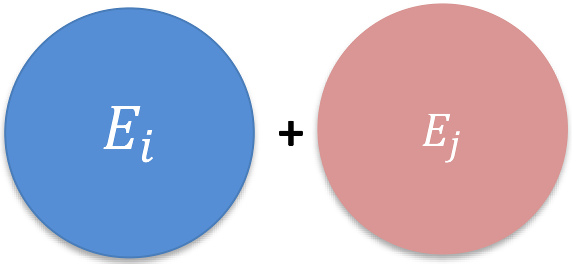
$$P(E_1) + P(E_2) + \ldots + P(E_n) = 1$$



Addition Rule

- for mutually exclusive events
  - The probability of the occurrence of either of two mutually exclusive events is equal to the sum of their individual probabilities.

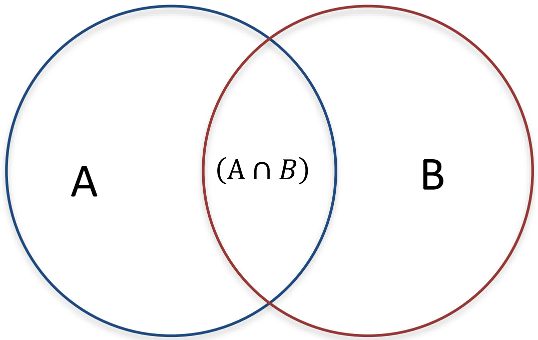
$$P(E_i + E_j) = P(E_i) + P(E_j)$$



Addition Rule

- for non mutually exclusive events
  - The probability that event A, or event B, or both occur is the sum of their probabilities minus the probability of their joint occurrence.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



Multiplication Rule

- Conditional Probability
  - The probability of A given B is the probability of the intersection of A and B divided by the probability of B

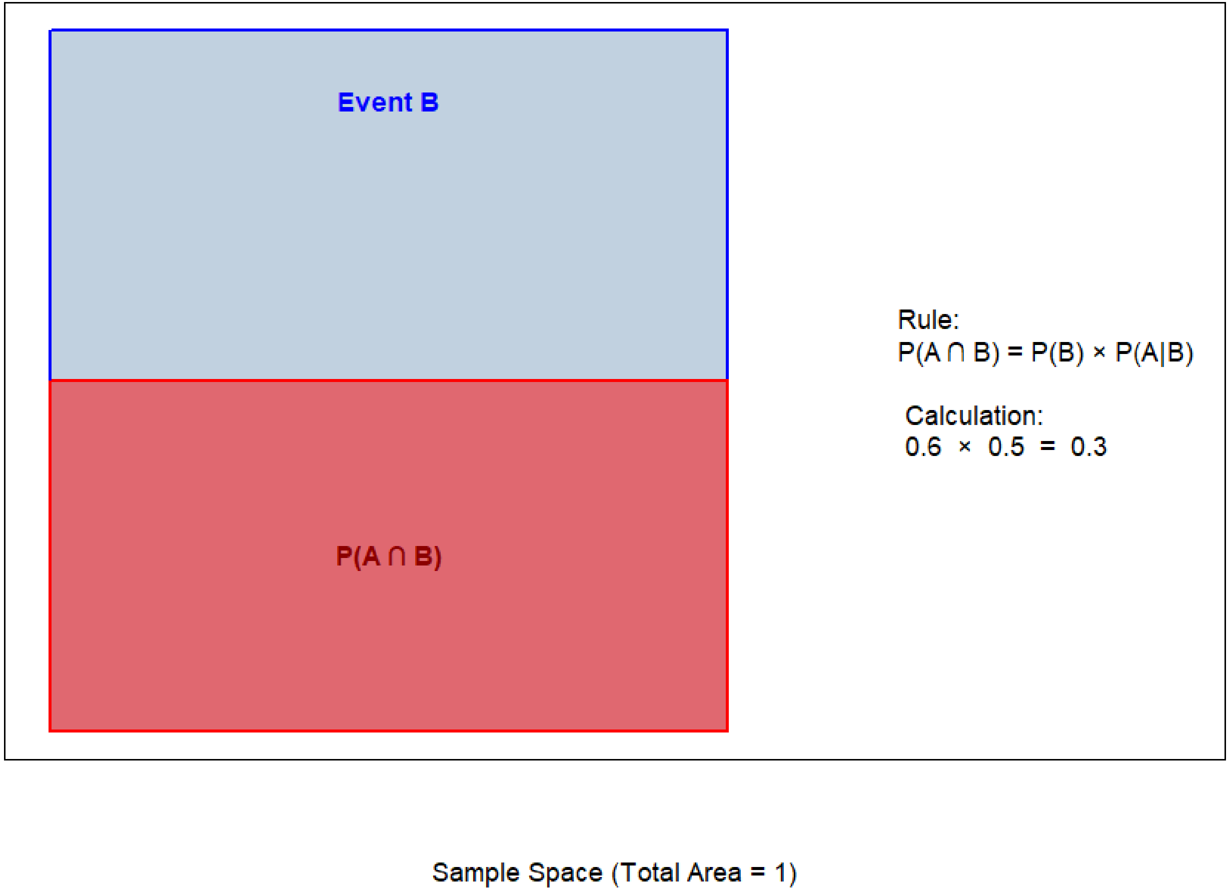
$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Multiplication Rule

- Joint probability
  - A joint probability is the product of a marginal probability and an appropriate conditional probability.

$$P(A \cap B) = P(B) \cdot P(A|B)$$

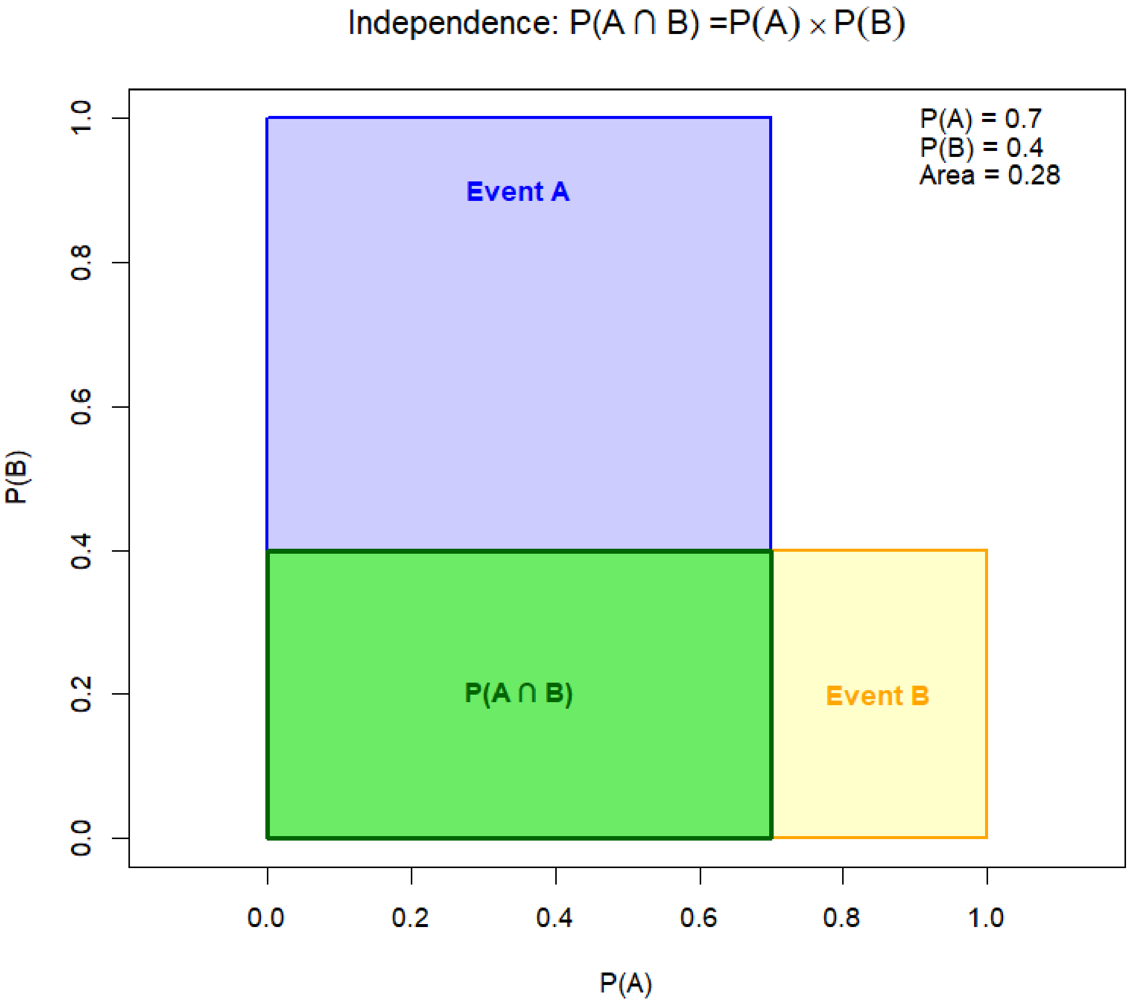
Visualizing the Multiplication Rule



Multiplication Rule

- Independence
  - The probability that event A, or event B, or both occur is the sum of their probabilities minus the probability of their joint occurrence.

$$P(A \cap B) = P(A) \cdot P(B)$$



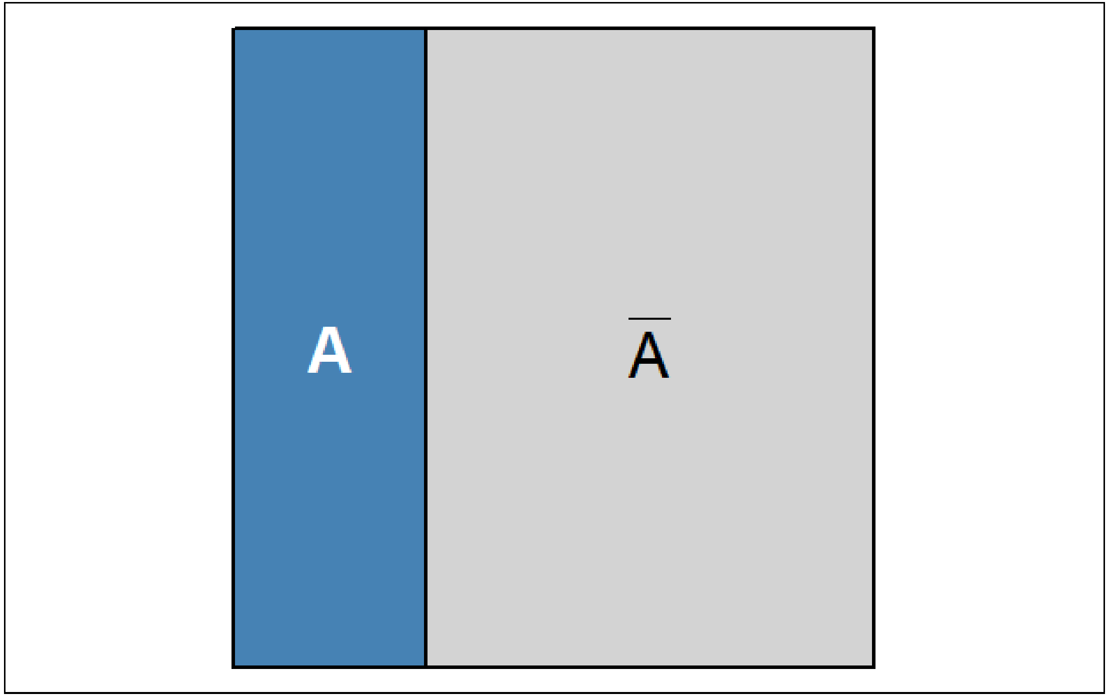
Independence: P(A ∩ B) =P(A) × P(B)
P(A) = 0.7
P(B) = 0.4
Area = 0.28
Event A
P(A ∩ B)
Event B
1.0
0.8
0.6
0.4
0.2
0.0
P(B)
0.0
0.2
0.4
0.6
0.8
1.0
P(A)

Complementary Events

- The probability of an event is 1 minus the probability of its complement.

$$P(\bar{A}) = 1 - P(A)$$

Complementary Rule: $P(\overline{A}) = 1 - P(A)$



P(A) = 0.3 P(not A) = 0.7

Total Probability Area = 1